

Data Flow Diagram

Date	1 October 2026
Team ID	SWTID-2026-2533
Project Name	EduGenie Google Gemini Powered Learning Assistant
Maximum Marks	2 Marks

Data Flow Diagram

A Data Flow Diagram (DFD) shows how data moves through the system - between external entities, processes and data stores. The symbol legend below explains the notation used in the EduGenie diagram.

DFD Symbol Legend

Symbol	Name	Description
	External Entity	A person or system outside EduGenie that sends or receives data (Student / Learner, Google Gemini API, Hugging Face Hub).
	Process	An activity that turns incoming data into outgoing data (Question Answering, Quiz Generation, Text Summarization).
	Data Store	A place where data is held for later use (Local Model Cache, Environment Variables).
	Data Flow	The movement of data between entities, processes and data stores, labelled with the data passed (for example: question, AI prompt). A dashed arrow is a fallback path.

EduGenie Level-1 DFD

The student uses the web page to send a question, topic or text. Request Handling (1) validates it and passes it to one of five feature processes (2 to 6). Processes 2 to 5 call the Google Gemini API; process 6 uses the locally stored LaMini-Flan-T5 model and falls back to Gemini. The result returns to the student through the same web page. EduGenie does not store any user data.

EduGenie Data Flow Diagram (Level 1)

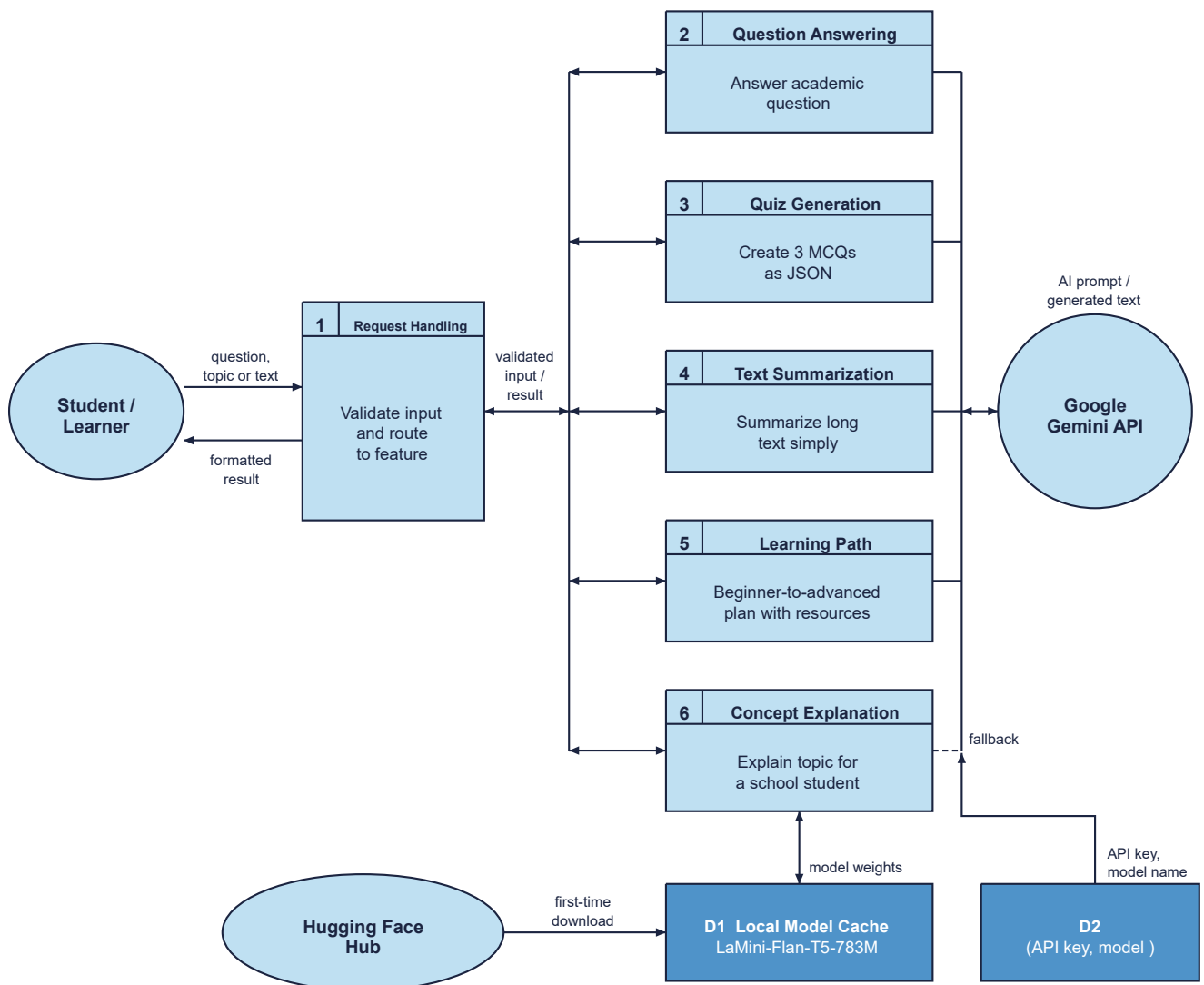


Fig. Data flow of the EduGenie learning assistant

Process Descriptions

ID	Process	What it does	Data in → out
1	Request Handling	FastAPI receives the request from the web page, rejects empty input with a clear error message and routes it to the right feature.	Question / topic / text → validated input; result → JSON response
2	Question Answering	Sends the student's question to Gemini and returns a short, accurate answer.	Question → AI prompt → answer
3	Quiz Generation	Asks Gemini for 3 multiple-choice questions in JSON, cleans and validates them (4 options, answer matches an option).	Topic / passage → quiz JSON
4	Text Summarization	Asks Gemini to summarize a long passage in simple language.	Passage → summary
5	Learning Path	Asks Gemini for a beginner, intermediate and advanced roadmap with estimated time and resources.	Topic → learning path
6	Concept Explanation	Explains a topic with the local LaMini-Flan-T5 model (loaded from D1); falls back to Gemini if the model cannot load.	Topic → explanation

Data Stores and External Entities

Name	Type	Purpose
D1 Local Model Cache	Data Store	Holds the LaMini-Flan-T5-783M model files downloaded once from Hugging Face and reused for every explanation.
D2 Environment Variables	Data Store	Holds the Gemini API key and model name (in a .env file locally, or in Vercel settings when deployed). Never stored in the code repository.
Student / Learner	External Entity	Sends questions, topics and text; receives answers, explanations, quizzes, summaries and learning paths.
Google Gemini API	External Entity	Receives AI prompts from processes 2 to 6 and returns generated text.
Hugging Face Hub	External Entity	Provides the LaMini-Flan-T5 model files on first use.